QR

QR RESTAURANT

Ordering & Loyalty

17 SCREENS • PRODUCT BLUEPRINT

QR menu, waiter-confirmed ordering, kitchen workflow, counter and online payments, digital stamps, AI menu import, and owner analytics.

SCAN | REQUEST | WAITER CHECK | CONFIRM | KITCHEN

Designed for dine-in restaurants, cafés, and food businesses

OFFLINE → ONLINE

5 problems. One system

Replace disconnected paper processes without removing the waiter from service.

Printed menus

Prices and availability become outdated; reprinting wastes time.

Manual ordering

Staff repeat orders, clarify choices, and calculate totals.

Shared QR misuse

A QR image can be forwarded; scanning cannot prove table presence.

Paper loyalty

Stamps are lost, duplicated, or forgotten.

Payment confusion

Online, counter, unpaid, and refunded orders get mixed up.

KEY DECISION

A customer order is a REQUEST first. A waiter verifies the physical table and confirms it. Only confirmed orders reach the kitchen.

THE SOLUTION

One app · six connected roles

Each role sees only the actions and data required for its work.

Customer

Scan, browse, request order, pay, track, earn stamps

Waiter

Verify table physically; confirm or reject requests

Kitchen

Receive confirmed orders; prepare and mark ready

Cashier

Collect counter payment; reconcile bills and receipts

Owner / Manager

Menu, staff, loyalty rules, reports, settings

Platform admin

Manage restaurant tenants and platform health

HOW IT WORKS

A QR order's journey

Waiter confirmation is the gate between customer request and kitchen preparation.

1

Scan table QR

Resolve restaurant and table context.

2

Browse and build cart

Choose items, variants, and quantities.

3

Submit request

Status: WAITING_FOR_WAITER; not confirmed.

4

Waiter verifies

Staff physically checks the stated table.

5

Waiter confirms

Only now does it enter the kitchen queue.

6

Prepare and serve

Kitchen updates progress; waiter serves.

7

Payment and loyalty

Reconcile payment; award eligible stamp.

SCREEN MAP

17 core screens

Grouped by the people who use them.

CUSTOMER

- 1 QR Landing
- 2 Digital Menu
- 3 Item Details
- 4 Cart & Request
- 5 Order Status
- 6 Loyalty Wallet
- 7 Profile & Review

STAFF

- 8 Waiter Requests
- 9 Kitchen Display
- 10 Cashier / Billing
- 11 Menu Scanner
- 12 Menu Editor

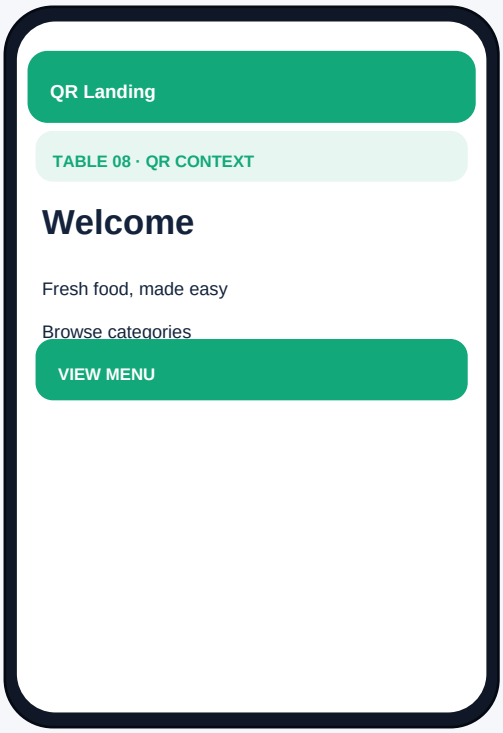
OWNER

- 13 Overview
- 14 Customers & Rewards
- 15 Reports
- 16 QR & Tables
- 17 Settings / Staff

Platform super-admin is a separate protected area for tenant administration; it is not part of restaurant staff navigation.

SCREEN 2 / CUSTOMER

QR Landing



PURPOSE

Open the correct restaurant and table context immediately after scanning.

WHAT THIS SCREEN MUST DO

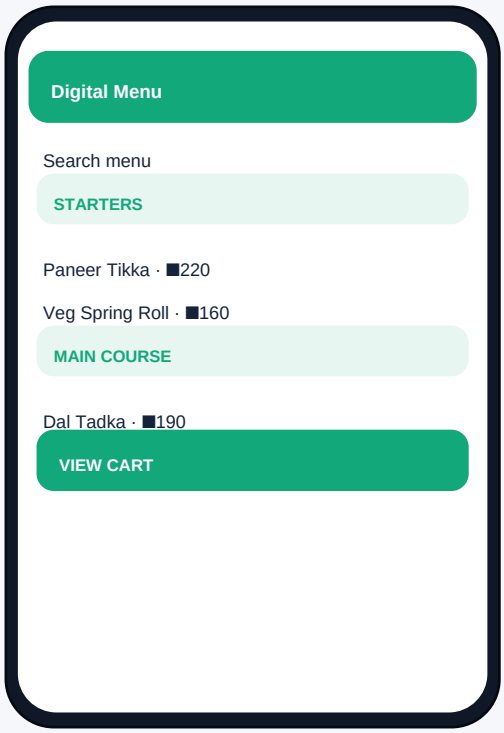
- Restaurant identity and opening status
- Detected table number
- Menu entry and offers
- Ordering paused/closed states

IMPORTANT RULE

A QR is not proof of physical presence. Waiter approval is mandatory for dine-in orders.

SCREEN 3 / CUSTOMER

Digital Menu



PURPOSE

Show current categories, item details, prices, and availability.

WHAT THIS SCREEN MUST DO

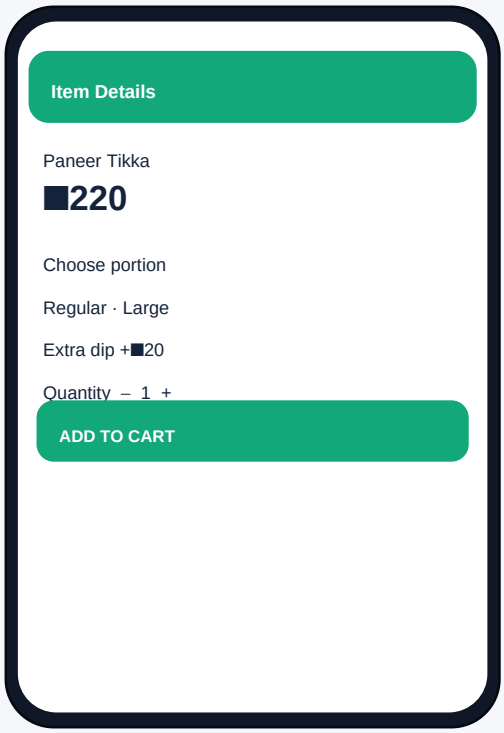
- Category tabs and search
- Item image, name, description, price
- Sold-out state and configured labels
- Persistent cart summary

IMPORTANT RULE

Revalidate prices and stock at checkout; never trust cached client totals.

SCREEN 4 / CUSTOMER

Item Details



PURPOSE

Choose valid variants, add-ons, quantity, and special instructions.

WHAT THIS SCREEN MUST DO

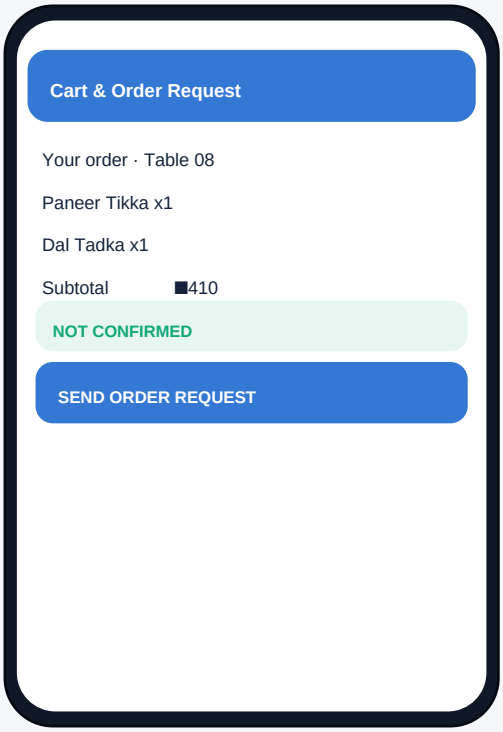
- Item description and current price
- Variants and add-ons
- Quantity and instructions
- Updated subtotal

IMPORTANT RULE

Server validates option IDs and prices; historical order snapshots remain unchanged.

SCREEN 5 / CUSTOMER

Cart & Order Request



PURPOSE

Review the cart and submit a request for waiter confirmation.

WHAT THIS SCREEN MUST DO

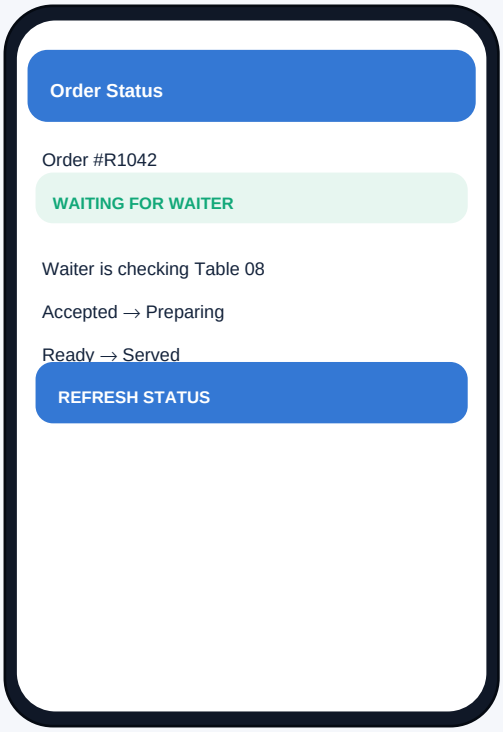
- Edit or remove items
- Subtotal, tax, charges
- Dine-in table context
- Submit request; show pending status

IMPORTANT RULE

Submitting creates a pending request only. It must not enter the kitchen queue.

SCREEN 6 / CUSTOMER

Order Status



PURPOSE

Show the real order state from request to serving.

WHAT THIS SCREEN MUST DO

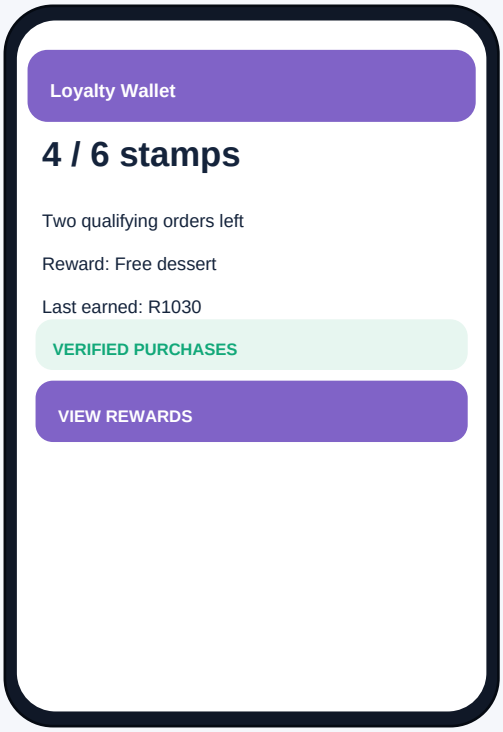
- Waiting for waiter verification
- Confirmed or rejected status
- Preparing, ready, served
- Payment status separately

IMPORTANT RULE

Never show CONFIRMED until the waiter's action is saved by the server.

SCREEN 7 / CUSTOMER

Loyalty Wallet



PURPOSE

Display verified stamps, reward progress, and redemption history.

WHAT THIS SCREEN MUST DO

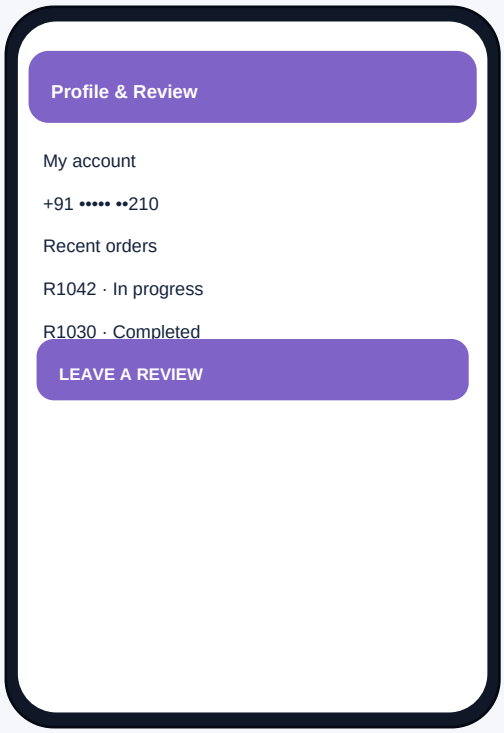
- Stamp progress and reward threshold
- Qualifying purchase rule
- Expiry and redemption history
- No automatic daily stamp

IMPORTANT RULE

A new day never creates a stamp by itself. Award only after a verified qualifying event.

SCREEN 8 / CUSTOMER

Profile & Review



PURPOSE

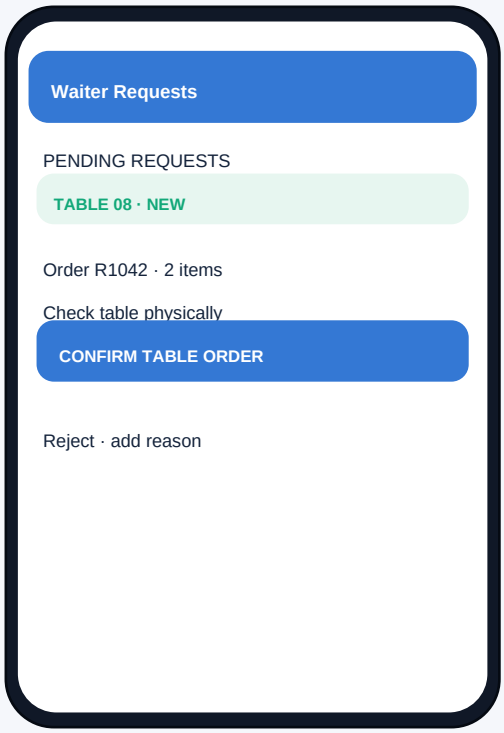
Access personal orders, receipts, loyalty, and feedback.

WHAT THIS SCREEN MUST DO

- Verified phone/profile
- Order history and receipts
- Loyalty activity
- Review completed order
- Privacy/data request route

SCREEN 9 / WAITER

Waiter Requests



PURPOSE

Show pending table requests so staff can physically verify and decide.

WHAT THIS SCREEN MUST DO

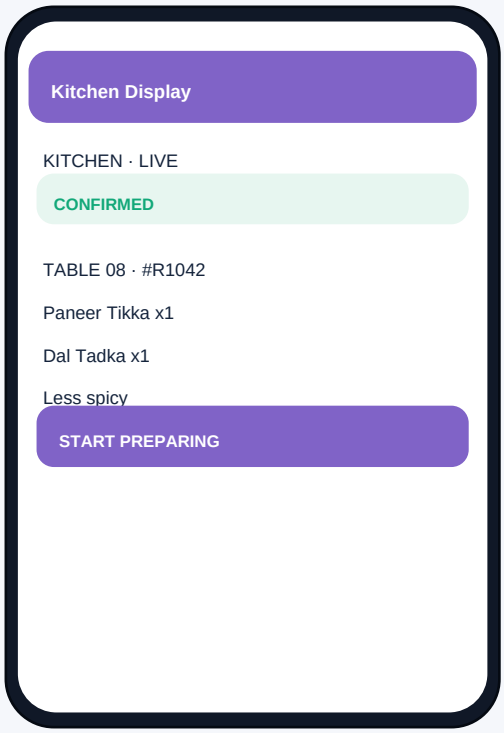
- New request alert
- Table number and order summary
- Physical table verification
- Confirm/reject with reason
- Log staff ID and timestamp

IMPORTANT RULE

Waiter confirms only after checking the customer is at the stated table.

SCREEN 10 / KITCHEN

Kitchen Display



PURPOSE

Display only orders confirmed by a waiter or authorized staff.

WHAT THIS SCREEN MUST DO

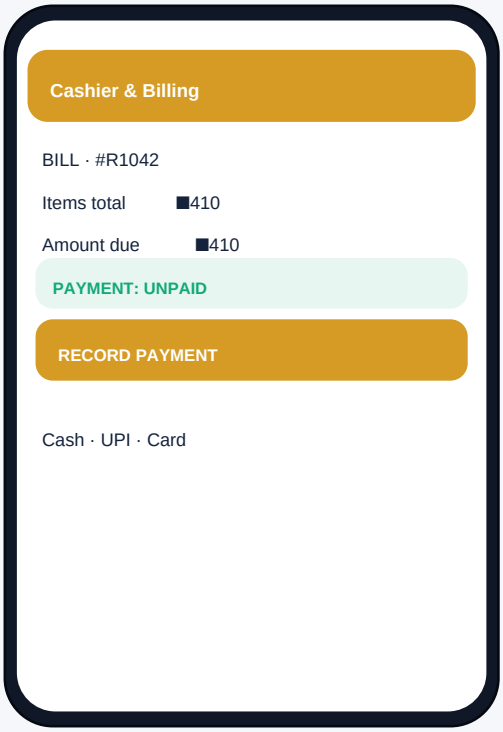
- Confirmed orders only
- Item modifiers and notes
- Preparation timer
- Accept/preparing/ready
- Reconnect recovery

IMPORTANT RULE

Requests waiting for waiter approval must never appear in the kitchen queue.

SCREEN 11 / CASHIER

Cashier & Billing



PURPOSE

Record counter payments and reconcile online transactions.

WHAT THIS SCREEN MUST DO

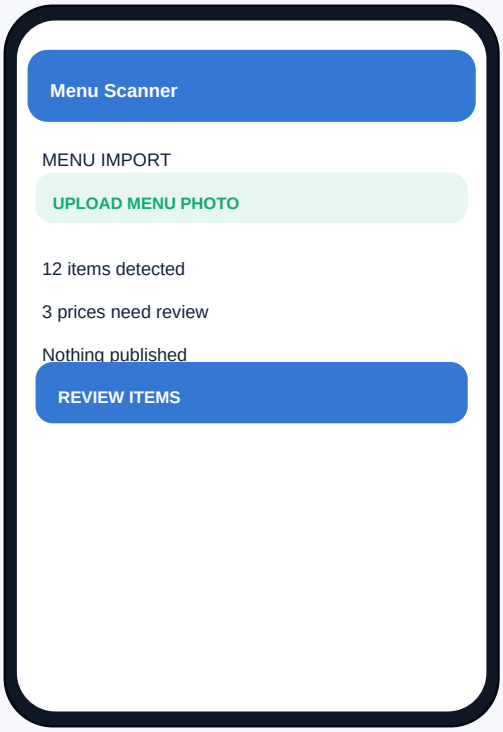
- Order and bill summary
- Cash, UPI, card methods
- Cashier identity and timestamp
- Online payment verification
- Refund and receipt history

IMPORTANT RULE

Order and payment states are separate. A browser redirect is not proof of payment.

SCREEN 12 / ADMIN

Menu Scanner



PURPOSE

Convert printed menus into editable draft items using OCR/AI.

WHAT THIS SCREEN MUST DO

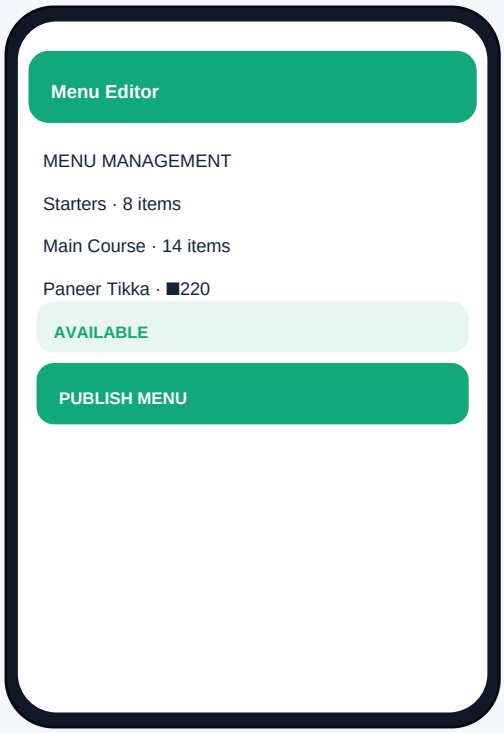
- Camera capture or PDF upload
- Extract category/name/price
- Highlight uncertain fields
- Human review of every price
- Import as draft, then publish

IMPORTANT RULE

AI must not invent missing prices. Human verification is required before publication.

SCREEN 13 / ADMIN

Menu Editor



PURPOSE

Manage categories, items, pricing, variants, and availability.

WHAT THIS SCREEN MUST DO

- Create/edit categories
- Update prices and images
- Variants and add-ons
- Sold-out/available controls
- Preview and publish

QR INTEGRITY

Outside the restaurant?

A static QR can be photographed or forwarded. Do not claim that QR sharing can be technically prevented.

1 · Table-specific QR

Each code maps to one restaurant and one table using a random, revocable identifier.

2 · Request remains pending

Scanning and submitting never confirms the order. It stays WAITING_FOR_WAITER.

3 · Waiter checks physically

Waiter verifies the customer is actually at the stated table before confirming.

4 · Optional stronger controls

Rotating QR, table PIN, or short-lived session token can add friction to QR sharing.

OUTSIDE THE RESTAURANT

Public menu browsing may remain available. Dine-in ordering requires a valid table context and waiter confirmation. Takeaway/delivery from outside should be a separate enabled flow.

CONNECTED BUSINESS RULES

Order, payment, and stamp states

Treat these as separate connected state machines.

ORDER STATUS

REQUESTED → WAITING_FOR_WAITER → CONFIRMED → PREPARING → READY → SERVED / COMPLETED. Rejected/cancelled paths require actor and reason.

PAYMENT STATUS

UNPAID / PENDING / PAID / FAILED / REFUNDED / PARTIALLY_REFUNDED. Online payment needs signed webhook verification.

LOYALTY STATUS

Eligible completed paid order → server checks rules/caps → ledger award. Refund follows documented reversal policy.

AUDIT EVENTS

Record request, waiter confirmation, kitchen update, cashier payment, refund, and stamp adjustment with actor/time.

Recommended rule: one stamp per qualifying completed paid order, with configurable minimum spend and daily cap. Use the restaurant's business timezone.

PRODUCTION FOUNDATIONS

Security, data, and traffic

These requirements are part of the product, not optional polish.

AUTHENTICATION

Secure staff login, session expiry/revocation, OTP attempt limits, password hashing, MFA for privileged users.

AUTHORIZATION

Server-side role checks, tenant isolation, customer order ownership, database row-level security.

TRANSMISSION

HTTPS/TLS, strict CORS, CSRF protection where relevant, schema validation, rate limits, upload limits.

PAYMENTS

Signed webhook verification, idempotency, immutable payment events, server-side totals, no card/CVV storage.

PERFORMANCE

Optimized images, pagination, indexes, safe caching, checkout revalidation, queue OCR and notifications.

RELIABILITY

Backups and restore tests, monitoring, health checks, alerting, retry/backoff, rollback plan.

ARCHITECTURE

One source of truth

Every client action passes through authenticated, validated server logic.



Tenant-scope every restaurant record. Store money as integer paise, timestamps in UTC, and display in restaurant timezone. Keep immutable order-line and payment event snapshots.

BUILD ROADMAP

Ship the reliable core first

Do not start with AI scanning or advanced analytics before transaction correctness.

PHASE 1 · CORE

Restaurant setup, staff login, QR/table mapping, menu, cart, waiter confirmation, kitchen queue, counter payment.

PHASE 2 · TRUST

Customer identity, stamp ledger, rewards, online payment, signed webhooks, refunds, audit history.

PHASE 3 · AUTOMATION

AI menu scanner, OCR review, notifications, reports, QR analytics, scheduled availability.

PHASE 4 · SCALE

Multiple branches, inventory/POS integration, advanced offers, platform billing, monitoring.

BEFORE LAUNCH

Testing checklist

Test the complete journey with real devices, staff roles, and payment sandbox events.

- QR opens the correct restaurant and table.
- Shared/outside QR request stays unconfirmed until waiter verifies table.
- Only confirmed orders appear in kitchen display.
- Waiter action is logged with staff ID and timestamp.
- Customer cannot access another customer's order.
- Restaurant A cannot read or modify Restaurant B data.
- Client cannot manipulate price, tax, total, payment, or stamp count.
- Duplicate order/payment/webhook/stamp events do not duplicate records.
- Counter payment records cashier, method, amount, and time.
- Refund/cancellation follows the loyalty reversal rule.
- OCR flags uncertain prices and never auto-publishes.
- Slow network, reconnect, upload failure, and pending payment paths work.

GO LIVE

Production release checklist

A successful build is not the same as a safe production launch.

- Separate development and production environments.
- HTTPS, domain, CORS, security headers configured.
- Secrets only in deployment environment.
- Database migrations reviewed and backup taken.
- Tenant isolation and permissions tested.
- Payment sandbox and signed webhook tests pass.
- Privacy, terms, cancellation/refund policy published.
- Owner and staff accounts created securely.
- QR codes physically tested on multiple phones.
- Cashier and kitchen staff trained.
- Monitoring, uptime checks, alerts enabled.
- Restore and rollback procedure tested.

DO NOT LAUNCH IF

Waiter approval can be bypassed, tenant data is exposed, payment can be forged/duplicated, stamps can be duplicated, or backups cannot be restored.

CLIENT DELIVERY

How the service is installed

A repeatable workflow for delivering the system to restaurants.

1

1 · DISCOVERY

Restaurant type, menu, branches, tables, order flow, tax, payment methods, loyalty rule.

2

2 · CONFIGURATION

Branding, categories, items, table QR codes, staff roles, business hours, policies.

3

3 · BUILD & TEST

Customer flow, waiter approval, kitchen, cashier, payments, loyalty, security, device tests.

4

4 · TRAINING

Owner, manager, waiter, cashier, and kitchen training using their real devices.

5

5 · LAUNCH & SUPPORT

Print QR codes, deploy, monitor launch, backups, maintenance, written support scope.

DEFINITION OF DONE

The connected system must work end-to-end

A screen is complete only when backend, permissions, persistence, and failure states work.

- Customer scans table QR and sees correct restaurant/table.
- Customer builds cart and submits a pending request.
- Waiter physically verifies table and explicitly confirms request.
- Only then does order appear in kitchen display.
- Kitchen updates preparation and ready status.
- Cashier records counter payment or gateway verifies online payment.
- Server awards eligible stamp exactly once.
- Owner can audit order, payment, and loyalty events.

CORE ACCEPTANCE RULE

No order is confirmed by scanning alone. A waiter must verify the table and confirm the request. The kitchen sees only confirmed orders. Payment and loyalty are independently verified.

ANTIGRAVITY BUILD INSTRUCTIONS

Build connected
flows, not mock screens.

Inspect the repository first. Report the existing stack, features, gaps, and risks. Then propose schema, API contracts, permissions, and routes before implementation.

For each phase: list files changed, explain data flow, implement the feature, add validation and error handling, write tests, run lint/type-check/build, and report what passed or failed.

NEVER MARK COMPLETE UNLESS

- ✓ UI and backend action are connected
- ✓ Authorization and tenant isolation are tested
- ✓ Persistence and failure states are implemented
- ✓ Tests pass; no fake buttons or hardcoded production data

Production reliability before extra features.

NEXT STEP

QR → request → waiter → kitchen

Start with the transaction backbone: table QR, customer request, waiter verification, confirmed kitchen order, payment reconciliation, and loyalty ledger.

RECOMMENDED MVP

- ✓ QR and digital menu
- ✓ Waiter approval queue
- ✓ Kitchen display
- ✓ Counter billing
- ✓ Verified stamp ledger

Reliable restaurant workflow first. Automation and scale second.